Program overview

19-May-2025 11:53

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Bachelor TW 2024	**Bachelor Technische Wiskunde 2024**						
1e jaar BSc TW 2024	**1st year BSc TW 2024**						
TW1-11	Problem Solving and Proofs	5					
TW1-12	Calculus	5					
TW1-13	Introduction to Programming	5					
TW1-21	Modelling 1	5					
TW1-22	Analysis 1	5					
TW1-23	Linear Algebra 1	5					
TW1-31	Discrete Mathematics	5					
TW1-32	Analysis 2	5					
TW1-33	Linear Algebra 2	5					
TW1-41	Modelling 2	5					
TW1-42	Ordinary Differential Equations	5					
TW1-43	Introduction to Probability	5					
2e jaar BSc TW 2024	**2nd year BSc Technische Wiskunde 2024**						
AM2010	Linear Algebra 2	6					
AM2020	Optimization	6					
AM2030	Ordinary Differential Equations	6					
AM2040	Complex Function Theory	6					
AM2050-A	Modelling 2A	3					
AM2050-B	Modelling 2B	3					
AM2060	Numerical Methods 1	6					
AM2070	Partial Differential Equations	6					
AM2080	Introduction to Statistics	6					
AM2090	Real Analysis	6					
keuzevakken 2e jaar BSc TW 2024	**Electives second year BSc TW 2024**						
AM2510	Decision Theory	6					
AM2520-H	History of Mathematics	6					
AM2520-P	Philosophy of Mathematics	6					
AM2550	Advanced Statistics	6					
AM2560	Applied Algebra: Codes	6					
AM2570	Markov Processes	6					
AM2580	Mathematical Models in Biology	6					
3e jaar BSc TW 2024	**3rd year BSc Technische Wiskunde 2024**						
AM3001	Bachelor Project	15					
AM3010	Bachelor Colloquium	3					
Minor (30 EC)							
keuzevakken 3e jaar BSc TW 2024	**Electives third year BSc TW 2024**						
AM3500	Mathematics Seminar	6					
AM3510	Mathematical Physical Models	6					
AM3530	Numerical Methods 2	6					
AM3540	Inverse Problems	6					
AM3550	Graph Theory	6					
AM3560	Advanced Probability	6					
AM3570	Fourier Analysis	6					
AM3580	Differential Geometry	6					
AM3590	Topology	6					

Year	**2024/2025**
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Education	**Bachelor Applied Mathematics**

Bachelor TW 2024	
Director of Education	Dr.ir. W.G.M. Groenevelt
Program Coordinator	E.M. van der Laag-Hoogendijk
Contact for students	You can find information on the degree program at https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/ Are you a first-year student and did not you find the answer to your question on the website? First contact your mentor. For higher-year students, the first person you may contact is the commissioner of AM (cow@ch.tudelft.nl). He or she knows answers to many questions, but can also refer to or ask other people from the degree program. During your entire studies, there are regular program meetings: In the first year there are quarterly meetings, in the higher years semester meetings. During these meetings you will find all the important information about the program. They are announced on Brightspace and are also in the schedule. For questions about help and guidance in case of personal problems or studies, all students can contact study advisors. Also check the website here: https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/organisatie/studieadviseurs/. For answers about specific subjects, first look at the Brightspace site of that subject and if you can not find the answer, contact the responsible teacher of that subject. Do you want to participate in improving the education? Then sign up for the student panels of your year. Contact your commissioner for education about this.
Program Coordinator	Bachelorcoordinator: Dr. B. van den Dries B.vandenDries@tudelft.nl
Term of validity	Please note! The information you find below is informative in nature and not binding! Binding information can be found in the rules and regulations (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). Keep an eye on Brightspace for changes and attend the program meetings for the latest information.
Program Goals	See Dutch version.
Program Structure 1	The BSc program Applied Mathematics lasts three years (3 x 60 EC). Each year is divided into 4 quarters of 10 weeks, where you follow courses worth of 15 EC. That means you have to spend 40 to 42 hours in your study every week. Of course, you may have to do more in one week than in another. It is therefore important to look ahead and plan your work. For students who started in 2024/25: The first year consists of 12 courses of 5 EC each. The second year also consists of 12 courses of 5 EC, of which two are elective courses. The third year differs considerably: In the first half you follow a minor (see below), in the second semester there are three elective courses of 5 EC, the last 15 EC you spend on your graduation project (also called a bachelor project). For students who started before 2024/25: The first year consists of 9 courses each of 6 EC and 2 courses of 3 EC. One of the courses of 6 EC is an elective course. The second year consists of 9 courses of 6 EC and 2 courses of of 3 EC. Also in the second year you have one elective course of 6 EC. The third year differs considerably: In the first half you follow a minor (see below), in the second semester there is 1 course of 3 EC and 2 elective courses of each 6 EC, the last 15 EC you spend on your graduation project (also called a bachelor project).
Program Structure 2	AM students and their teachers are part of the academic community of TU Delft. Each community has rules of conduct and TU Delft has laid down this in an ethical code (Code of Ethics. The following rules come from this code: Teachers and students: - have respect for others; - act involved, transparent and integer; - contribute to an inspiring work and study environment; - trust each other and avoid conflicts of interests; - respect each other's property and resources and those of the university Teachers: - behave righteously and respectfully with each other and students - aim for high quality and always strive for improvement; Students: - be fair and respectful towards each other and university staff; - get the best out of themselves by actively participating in their education and extra curricular activities; - encourage each other and their teachers by asking analytical questions and conducting well-argued discussions;
Exam requirements	The exam requirements state when you qualify for a AM diploma. You will find them in the Rules and Regulations of the Board of Examinors (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/). In summary, you get a AM diploma if you have completed all the courses of the program, including the minor, with as a result a V (sufficient) or a grade of 6 or higher. Your must apply for your diploma through the form on the forms website: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/forms/.
With Honours Regulation	Students who complete the program AM (3 years), have an minimum average of 8 for all courses, and have a minimum of 8 for the bachelor project, receive their diploma with the mark 'with honours' (also: cum Laude). The exact requirements can be found in the Rules and Guidelines of the Board of Examinors (https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/regulations/).
Minor Semester	In the first half of your third year you follow a minor. The minor is meant to give you the opportunity to deepen, broaden your knowledge and to obtain a foreign experience. More information can be found at minors.tudelft.nl.

Gives access to	The bachelor AM gives access to TU Delft Master Applied Mathematics. You also have access to the masters of the same name at the University of Twente and Eindhoven University of Technology. To be admitted to the mathematics master's programs from other Dutch and foreign universities you may need to do a switching program. With a switching program, it is also possible to be admitted to non-Mathematics Master programs.
Expected prior Knowledge	The admission requirements for the AM program are: https://www.tudelft.nl/en/education/programmes/bachelors/tw/bachelor-of-applied-mathematics/admission-requirements/ The AM program advises students, who scored less than 8 for Maths B and 7 for English, NOT start the program. In recent years, it appears that these students have insufficient knowledge of English and Mathematics to complete the courses of the first year.
Education Methods	There are many different types of education at AM. The teachers try to convey the material in the most appropriate way. The most common forms of education are briefly described below: - Class Lecture: At a class lecture a teacher gives a lecture in a classroom. Often this is supplemented with homework assignments. - Instruction: An instruction is an exercise class. Under the supervision of a teacher and / or student assistants, you will work on assignments. Usually you are in an instruction with only a part of your fellow students in one room, so the group does not get too big and you get the chance to ask questions. - Colstruction: A colstruction is a combination of a class lecture and instruction. Alternating, the teacher will explain and you work by yourself or groups. - Practicum: During a practical session you (usually self-employed) do programming assignments and practical exercises under the guidance of student assistants. - Project: A project is a practically oriented exercise where you make a big assignment in a group. Usually, something has to be delivered at the end of the project. Working together in the group is an important part of the project. At AM you will encounter this form of work at the Modeling courses in the first and second years. - Self-study: this seems a stranger in the midst, but it is indeed the most important form of education! It is important that you get started yourself by all forms above. Only in this way make sure that you can apply everything that you have been taught by yourself.
Honours track	For good students looking for additional challenges, there is the Honors Program Bachelor (HPB) from the second year. It is accessible to all students who completed the first year in one year and had a minimum of 8 for all subjects on average. Students with an average of 8 can apply for admission. More information can be found here: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/honours-programme/

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

1e jaar BSc TW 2024	
Program Coordinator	E.M. van der Laag-Hoogendijk
Program Coordinator	Bachelorcoordinator: Bart van den Dries BC-TW@TUDelft.nl
Prerequisites	There are no entry requirements for study components of the first year.
Introduction 1	
Introduction 2	
Introduction 3	As with all bachelor programs at TU Delft, at Applied Mathematics there is a norm of 45 EC for a binding study advice (BSA). For more information, see http://bsa.tudelft.nl.
Introduction 4	

TW1-11	Problem Solving and Proofs	5
Responsible Instructor	Dr. B. van den Dries	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch (on request English)	
Course Contents	see information in Dutch	
Assessment	written	
Co-Instructor	Dr.ir. W.G.M. Groenevelt	

TW1-12	Calculus	5
Responsible Instructor	Dr. F.J. van de Bult	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch (on request English)	
Course Contents	This is a Dutch course, as such you should look at the Dutch version of the studyguide.	
Assessment	written	
Co-Instructor	Y. Murakami	

TW1-13	Introduction to Programming	5
Responsible Instructor	Drs. P.R. van Nieuwenhuizen	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Co-Instructor	Drs. I.A.M. Goddijn	

TW1-21	Modelling 1	5
Responsible Instructor	Prof.dr.ir. M.B. van Gijzen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch (on request English)	
Course Contents	This course is the first of a set of four mathematical modelling courses. These modelling courses teach how to use mathematical tools to analyse real-life problems. In this first course the modelling cycle is introduced and skills from calculus 1 and programming are used to define, analyse and improve mathematical models based on ODEs. The students also get an introduction into scientific literacy, writing and presenting.	
Study Goals	By the end of the course, the student should be able to: - apply the modelling cycle (problem definition - formulate mathematical model - calculations - validation) to an ordinary differential equation (ODE) problem - write a computer program to solve the ODE numerically, perform numerical experiments and assess the outcomes - work in a small group and efficiently interact with the colleague and supervisor. - find and consult scientific literature related to the research project. - present the work to her/his peers by means of a poster - report their modeling outcomes in writing (using LaTeX)	
Education Method	Practical in student teams of two, with a tutor. Scientific writing with formal meetings and assignments. Attendance is mandatory for all scheduled course activities.	
Assessment	Communication with tutor, report, and poster	
Judgement	The project is evaluated per group. The performance during the lab sessions, the written report and the poster are the input for the grade. The quality of the research counts for 40 % of the grade. The report counts for 25 %, the performance for 25% and the poster for 10%. The introduction into scientific literacy and the scientific writing part of the course have to be concluded with a 'Pass".	
Co-Instructor	Dr.ir. M. Keijzer	

TW1-22	Analysis 1	5
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	This course builds upon the courses Problem Solving and Proofs (TW1-11) and Calculus (TW1-12) and their contents are closely linked and interwoven.	
Course Contents	This is an introduction to analysis for functions of one real variable. The following topics are treated: - sequences - series - continuity - differentiation - integration.	
Study Goals	At the end of the course, you should be able to: LO1) restate the definitions and give examples of the notions covered during the course LO1.1: convergent, bounded, divergent and Cauchy sequence, subsequence, supremum, infimum, maximum, minimum, (strictly) monotone sequence, limes superior and inferior LO1.2: convergent and absolutely convergent series LO1.3: continuous, bounded and uniformly continuous function, continuous limit LO1.4: differentiable function, (strictly) monotone function LO1.5: Riemann integral, uniform convergence, improper integral LO2) restate the statements of the results covered in the lectures on the notions listed in LO1 and explain their proofs LO3) apply the definitions together with the theorems treated during the course to prove results for sequences, series, continuous functions, differentiable functions, integrals and sequences of functions	
Education Method	The course consists of frontal lectures where the theory is explained. These are complemented with exercise classes.	
Reader	Lecture Notes Analysis 1 (available on Brightspace).	
Assessment	There are two written partial exams. The midterm exam takes 1 hour and counts 25% of the final grade while the final exam takes 3 hours and counts 75% of the final grade. The resit exam fully determines the final grade (the partial exams will not contribute to the final grade in this case). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	No course materials are permitted.	
Judgement	There are two ways to obtain a final mark for the course Analysis 1, TW1-22: 1. The final grade is determined by the grades of the partial exams, the first counting 25% and the second counting 75% of the final grade. OR 2. The final grade is the grade of the resit. Earlier partial exam results do not count towards the final mark. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

TW1-23	**Linear Algebra 1**	5
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	0/4/0/0 HC - 0/4/0/0 WC	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Required for	Many courses later on will use concepts and techniques of linear algebra 1, especially analysis 2, introduction to probability, algebra 1, ordinary differential equations, partial differential equations, numerical methods, systems theory, introduction to statistics, optimization, complexe function theory, lineare algebra 2, applied algebra, markov processes.	
Course Contents	Vectors are well-known: they are often represented as arrows that indicate a magnitude and direction, and they can be added and/or multiplied by a scalar. The subject of linear algebra studies vectors. Initially, a different representation of the arrows is treated, making calculations with vectors simple. One of the advantages of this representation is that you realize that this method of calculation also occurs in other situations. For example, functions can also be seen as vectors, and calculations can be made with, for example, complex numbers. This leads to the concept of (real, complex) vector spaces. We will study when and how vector spaces (or subspaces of vector spaces) can be described with a finite number of vectors. Central concepts here are independence, basis, and dimension. The means to calculate these is by solving systems of linear equations. To relate vector spaces (subspaces) to each other, you use functions from one vector space to another. The ones interesting to linear algebra are linear mappings. We will see that linear mappings between finite-dimensional vector spaces can be represented as matrices. Calculations can be performed with matrices: addition, scalar multiplication, multiplication, and transposition are operations you will need to master. You also need to know what it means for matrices to be invertible, how to calculate the inverse of a matrix, and what this has to do with the corresponding linear mapping. You will also learn how the different matrix representations of a linear mapping are related. Another concept (which has both a geometric and algebraic interpretation) is the determinant of a square matrix. This, for example, is a measure of the volume scaling factor in linear mappings, but it also indicates whether a matrix (or linear mapping) is invertible or not. In the course Linear Algebra 2, the theory is further expanded. Linear algebra has many applications in both fundamental mathematics (algebra, probability theory) and applied mathematics (differential equations, numerical mathematics, statistics, systems theory) and physics (quantum mechanics). It is therefore one of the important foundational courses for the program.	
Study Goals	These can be found on the brightspace-pages of the course Linear Algebra 1 and in the dutch version of this site.	
Education Method	14 lectures, 14 instructions. For a full description see the dutch version.	
Literature and Study Materials	Book: Linear Algebra 5th edition written by Friedberg, Spence en Insel (publishe by Pearson). Lecture slides (they will be published on brightspace). If there are: Extra exercises (they will be published on brightspace).	
Assessment	2 partial exams (written): in week 5 of Q2 two hours with weight 40%; in week 9 or 10 of Q2 two hours with weight 60%. The final grade is the weighted average of the two unrounded grades. Condition: the unrounded grade of the 2nd partial exam must be at least 4.0. Otherwise the final grade is the minumum of the grade of the 2nd partial exam. and the weighted average of the two unrounded grades. The resit (written) takes 3 hours and has a weight of 100%. It will be scheduled somewhere in Q3 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. J.H. Weber	

TW1-31	Discrete Mathematics	5
Responsible Instructor	Dr. C.E. Groenland	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch (on request English)	
Required for	Probability, Optimalisation, Graph Theory	
Expected prior knowledge	From Problems and Proofs: notion of sets (complement, intersection, union), proof by induction.	
Course Contents	This course provides an introduction to discrete mathematics. We study discrete objects such as integers, sets, and graphs. Examples of graphs include social networks or road networks. In the first half of the course, we delve into various counting principles, such as binomial and multinomial coefficients, double counting, inclusion-exclusion, and recurrence relations. We also cover properties of integers such as modular arithmetic and the uniqueness of prime factorization. In the second half of the course, we explore basic concepts of graph theory such as Eulerian walks, trees, matching, planar graphs, and graph coloring. Throughout the course, emphasis is also placed on applying 'discrete thinking' to solve problems and on algorithmic aspects, including Euclid's algorithm, breadth-first search, bipartite matching, and greedy algorithms.	
Study Goals	1) Use the language of discrete mathematics to formalise mathematical and real-life problems. 2) Formulate rigorous mathematical proofs. 3) Solve counting problems using techniques such as double counting, binomial coefficients, bijections, inclusion-exclusion and recurrence relations. 4) State the main definitions and theorems and explain the proofs of these theorems. 5) Prove new mathematical statements by combining concepts, examples and theorems covered in the course. 6) Apply algorithms from the course such as Euclidean's algorithm for finding the greatest common divisor, augmenting path algorithm for finding a maximum matching and tree algorithms such as depth-first search.	
Education Method	4h lectures and 4h tutorials/instructions per week	
Books	Discrete Mathematics Elementary and Beyond from L. Lovász, J. Pelikán and K. Vesztergombi	
Prerequisites	Problems and Proofs	
Assessment	Written exam (midterm 40%, endterm 60%)	
Co-Instructor	Prof.dr. D.C. Gijswijt	

TW1-32	Analysis 2	5
Responsible Instructor	Dr. M.P.T. Caspers	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	Dutch (on request English)	
Expected prior knowledge	You will certainly have the required prerequisites if you have followed the courses Analysis 1, Calculus 1, and to a certain degree also Linear Algebra 1 and Problems & Proofs. If you have not passed these courses yet you can still enter Analysis 2 but material covered in those courses will be assumed to be known. More precisely we assume that students are familiar with: Analysis *Sets, functions, real numbers. *Epsilon-delta arguments, limits and continuity. *Formal notions of differentiation of single variable functions. *Taylor approximation of a single variable function. *Integration of single variable functions, primitives and the Fundamental Theorem of Calculus linking primitives to integration. Linear Algebra *Matrices, the determinant and the n-dimensional vector space R^n. Calculus *Partial derivatives on a computational level *Gradient of a function *Integration over 2- and 3-dimensional regions in R^2 and R^3 of an R-valued function, the fact that such integrals decompose as repeated single variable integrals, and the most common coordinate changes such as polar, spherical or cylindrical coordinates (the latter will be repeated in the course in a larger generality). Students should know how to compute such integrals and interpret the outcome.	
Course Contents	We treat the following material. *Lecture notes "Analysis 1" by Jan van Neerven: Sections 6.1, 6.2, 6.3, 6.4. *Book by Lax and Terrell: Sections 2.1, 2.2, 3.5, part of 6.6 (and maybe 6.4), 7.1, 7.2, 7.3, 8.1, 8.2, 8.3. Concretely this covers the following topics. The following topics will be covered on an analytic level, meaning that students should understand both the proofs as well as applications of the theory: *Power series *The exponential function *The logarithm *Trigonometric functions *Functions of several variables, vector fields, curves *Closed and open subsets of R^n, continuity and limits of functions of several variables. Note that this material comes both from the lecture notes and the book. The following topics will be covered on a calculus level, meaning that students should be able to apply to material (typically to determine integrals) and understand the meaning of the concepts, but need not to know the proofs. *Gradient, divergence, curl *Substitution rule and change of variables (may restrict to 2 or 3 variables, to be decided later) *Line integrals. *Conservative vector fields. *Surface integrals *Green, Stokes Gauss theorem (divergence theorem)	
Study Goals	*The student is able to analyze and apply the theory of power series *The student is able to analyze and apply the exponential function *The student is able to analyze and apply the logarithm *The student is able to analyze and apply the basic trigonometric functions *The student is able to analyze and apply the theory of functions of several variables, vector fields, curves. *The student is able to analyze closed and open subsets of R^n, continuity and limits of functions of several variables. *The student can apply the gradient, divergence, curl and compete these for concrete functions. *The student can apply the substitution rule and change of variables (may restrict to 2 or 3 variables, to be decided later) to compute integrals of multivariable functions. *The student understands what line integrals are and can apply this to determine the line integral of a concrete functions. *The student understands what a conservative vector field is and can apply this to determine integrals over closed paths. *The student can apply surface integrals to compute the integrals of functions over surfaces *The student can apply the theorems of Green, Stokes and Gauss (the latter also being known as the divergence theorem) to determine integrals of concrete functions.	
Education Method	*Lecture notes by Jan van Neerven "Analysis I" (Section 6). *The book by Peter D. Lax and Maria Shea Terrell, "Multivariable calculus with applications", Springer 2018. There is a digital version of the book that is available for free and which can also be used to follow the course.	
Assessment	*There is a written mid-term exam, call the grade M. *There is a written end-term exam, call the grade E. *The total grade is 40% * M + 60% * E. *You cannot finish the course by only taking the end-term exam and not the mid-term. *There is a written retake exam, call the grade R. *If you take the retake exam, then R is your grade. I.e. the midterm does not count in your final grade. In particular, for the retake it is not necessary to take the mid-term.	
Co-Instructor	Dr. S.E. Zegers	

TW1-33	Linear Algebra 2	5
Responsible Instructor	Dr.ir. S. Jain	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	Fundamental building block for many other courses like Analysis 2, Probability theory, Algebra 1, Ordinary differential equations, Partial differential equations, Numerical methods, Systems theory, Optimization, Complex Analysis, Applied algebra, functional analysis.	
Expected prior knowledge	Linear algebra 1: TW1-23 or equivalent	
Course Contents	This is the second half of the course covering fundamental aspects of Linear Algebra in BSc applied mathematics program (first year). Specifically, this course deals with the following topics: + Spectral analysis: eigenvectors and generalized eigenvectors of linear operators, eigenvalue decomposition (diagonalization) and the Jordan canonical form of a linear operator. + Orthogonalization: Inner products, adjoints, orthogonal projections, orthogonal bases and their properties, Gram-Schmidt procedure. + Optimization: least squares, constrained quadratic optimization, Lagrange multipliers. + Singular value decomposition, low-rank approximations, pseudo inverses. + Computational aspects of the above concepts. The topics covered are important for many fields in applied mathematics, including differential equations, data science, and numerical analysis.	
Study Goals	At the end of the course, the student will be able to: 1. Perform spectral analysis and diagonalization of linear operators using the notion of eigenvalues, eigenvectors and Jordan canonical forms. 2. Construct orthogonal bases for inner product spaces using the Gram-Schmidt procedure. 3. Solve least squares using adjoints and solve constrained quadratic optimization problems using Lagrange multipliers. 4. Perform singular value decomposition (SVD) to compute pseudo-inverse and low-rank approximations of a linear operator. 5. Develop a basic understanding of the numerical and computational aspects of the above mathematical concepts.	
Education Method	In weeks 1, 2, 3, 4, 6, 7, 8 there will be two two-hour lectures and two two-hour instructions. The instruction will be generally about the topics covered in the previous lecture.	
Literature and Study Materials	Linear Algebra (5th edition) by Stephen Friedberg, Arnold Insel and Lawrence Spence, Pearson.	
Assessment	The final grade of this course is determined by two tests; one in week 3.5 and one in week 3.9 or week 3.10. After the final test, the final grade for this course is calculated as follows: (grade test 1)/3 + 2*(grade test 2)/3 The course is passed if this grade is 5.75 or higher AND the grade of the second test is 5 or higher. There will be one overall resit opportunity somewhere in Q4, which counts towards 100% of the final grade. It is not possible to do a resit for separate tests. If the grade for the resit is higher than the final grade, it will replace the final grade.	
Co-Instructor	Dr. A. Heinlein	

TW1-41	Modelling 2	5
Responsible Instructor	Dr.ir. M. Keijzer	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch (on request English)	
Course Contents	This course is the second of a set of four mathematical modelling courses. These modelling courses teach how to use mathematical tools to analyse real-life problems. In this course, students use the modelling cycle to solve optimisation problems. The students have to use their programming skills, and further practise scientific writing and presenting. Moreover, students will learn team work skills.	
Study Goals	By the end of the course, the student should be able to: - Apply the modelling cycle (problem definition - formulation of a mathematical model - calculations - validation) to an optimization problem - Write a computer program using specific optimization software - Demonstrate basic teamworking skills - Report their modelling outcomes verbally and in writing (using Latex) - Critically reflect on the team as a whole, their own roles in the team and that of the other team members	
Education Method	Practical in student teams of four, with a tutor. Team work with formal meetings, minutes and peer review. Team assignments to practice Team work skills.	
Assessment	Formative assessment: - Peer feedback by group members (2x) - Peer feedback on the draft report of another team Summative assessment: S1: A written report on the research project for each team: 40% of the grade for the contents, 20% of the grade for the form of the report, S2: A poster and pitches for each team at the poster presentation: 10% of the grade, S3: An evaluation report written by each team + communication with tutor + control over process, 30% of the grade S4: Individual reflections on groupwork assignments: pass/fail Repair possibility: In case of insufficient results a repair option exists for the research report and the reflection assignments, in accordance with Article 2, Examination requirements, Clause 4 of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. J.T. van Essen	

TW1-42	Ordinary Differential Equations	5
Responsible Instructor	Dr. A. Geyer	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course is about ordinary differential equations. For first order equations we will treat linear, separable, and exact equations. We will prove existence and uniqueness of solutions, using Picard iterations. For second order equations we will treat linear equations, power series solutions, and the method of Frobenius. We will treat systems of linear differential equations and discuss stability and phase plane analysis.	
Study Goals	After successfully completing the course, you will be able to use the theory of ordinary differential equations (ODEs) to solve or perform qualitative analysis on (systems of) differential equations. In particular, you can 1) solve linear (systems of) ODEs 2) solve separable ODEs and exact ODEs 3) determine the stability of solutions of ODEs 4) use theorems and techniques introduced in the course to prove properties such as local existence and uniqueness of solutions. 5) Use ideas and techniques from the proofs in the course to prove related results. 6) solve second order ODEs using power series and Frobenius theory. 7) determine periodic or asymptotic behaviour of solutions of ODEs. 8) sketch the phase planes of nonlinear systems of ODEs.	
Education Method	Lectures (4 hours a week) and exercise classes (4 hours a week)	
Assessment	written	
Co-Instructor	Dr. Y. van Gennip	

TW1-43	Introduction to Probability	5
Responsible Instructor	Dr.ir. R. Versendaal	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	see description in Dutch	
Assessment	written	
Co-Instructor	J. Komjáthy	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

2e jaar BSc TW 2024

Program Coordinator	Bachelorcoordinator: Bart van den Dries BC-TW@TUDelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the courses AM2050-A Modeling-2A (simulation) and AM2050-B Modeling-2B (differential equations), it is mandatory to have finished the courses AM1050-A Modeling-A and AM1050-B Modeling-B.
Introduction 1	Enrollment: For the courses AM2050-A Modeling-A and AM2050-B Modeling-B you have to enroll in Brightspace beforehand and choose a topic. For the (partial-)exams of all courses, you have to register via OSIRIS ultimately two weeks before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/. If you do you not want to participate in a study component for which you have enrolled: Please unenroll yourself via Brightspace/OSIRIS!
Introduction 2	Special components year 2: In the third quarter you will have an elective course. You can choose from all elective courses that are listed in the study guide (courses from Leiden included), but the schedule of only the courses with a AM2-code have been aligned with the schedule of the other second year courses. To obtain your TW-diploma you need at least one elective course with a AM3-code, or an elective course from Leiden.
Introduction 3	Outlook year 3: In the third year you will do a minor in the first semester. You will choose this minor already in your second year. For more information, see http://minors.tudelft.nl. Because the content of minors may overlap with the content of the bachelor program Applied Mathematics, you need to have the minor of your choice approved beforehand by the bachelor coordinator Joost de Groot. Do you want to go abroad in your third year? Then start to inform yourself at https://www.tudelft.nl/studenten/ewi-studentenportal/organisatie/international-internship-office.

AM2010	Linear Algebra 2	6
Responsible Instructor	Dr. A. Heinlein	
Contact Hours / Week x/x/x/x	4/0/0/0 hc: 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030)	
Course Contents	The course consists of two parts: The first part of the course focuses on the concept of vector spaces and their applications. Topics to be discussed are: basis transformations, quotient spaces, linear operators and their properties, subspaces, eigenvalues and eigenvectors, the Jordan canonical form, inner product and normed vector spaces and their operators, projections, and quadratic forms. The second part of the course addresses concepts from numerical linear algebra. Topics to be discussed are: special matrix types, vector and matrix norms, orthogonal decompositions and orthogonalization via the Gram-Schmidt process, Gershgorin's circle theorem for estimating eigenvalues, as well as the singular value decomposition and pseudo inverses of a matrix.	
Study Goals	At the end of the course, you should be able to: + Construct finite dimensional vector spaces, subspaces, and inner product spaces as well as linear operators on those spaces. + Analyze finite dimensional vector spaces, subspaces, inner product spaces, and linear operators with respect to their properties via derivations and proofs. + Compute eigenvalues and (generalized) eigenvectors of linear operators respectively perform their (partial) diagonalization. + Compute the singular value decomposition of matrices and linear operators.	
Education Method	The course consists of lectures and practical sessions. The theoretical concepts will be explained during the lectures and illustrated by examples. The practical sessions are meant for reviewing the theoretical concepts and for applying them to examples, which is a good preparation for the exam.	
Computer Use	Basic experience with MATLAB (or Python) is recommended for checking the correctness of some computations of, e.g., eigenvalues/vectors, Jordan canonical forms, and singular value decompositions.	
Literature and Study Materials	Lecture slides will be provided	
Books	L. Sadun, Applied Linear Algebra. The Decoupling Principle, Second Edition, AMS, 2008.	
Assessment	A three-hours written exam counting for 100% of the final grade. In case of an insufficient result, a three-hours resit exam is offered during the third quarter counting for 100% of the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Non-programmable pocket calculator	
Co-Instructor	Dr. M. Möller	

AM2020	Optimization	6
Responsible Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	0/4/0/0 hc 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra 1 (AM1030), Analysis 2 (AM1070), Mathematical Structures (AM1010)	
Course Contents	Introduction to optimization, convexity. Modelling (Mixed) Integer Linear Programming problems. Linear programming: Simplex method, duality. Network optimization: shortest paths, max flow. Integer linear programming: branch & bound, Gomory cutting planes. Nonlinear optimization: Newton method, steepest descent. Complexity: NP-hardness. Approximation algorithms.	
Study Goals	The students are able to formulate discrete optimization problems as (mixed integer) linear programming problems. The students are able to execute basic algorithms for solving linear programming, integer linear programming and nonlinear programming problems, to understand the outcomes of the algorithms and to apply duality. The students know basic theorems of linear and nonlinear optimization and are able to prove variations of these theorems. The students can apply basic algorithms for network optimization problems (shortest path, max flow, travelling salesman problem). The students are able to design reductions between optimization problems and to prove correctness of those reductions.	
Education Method	4 hours lecture and 4 hours of exercise class per week, plus feedback exercises and homework exercises	
Literature and Study Materials	Lecture notes	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	3 hours	
Permitted Materials during Tests	A non-graphical calculator	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr.ir. L.J.J. van Iersel	

AM2030	Ordinary Differential Equations	6
Responsible Instructor	Dr. Y. van Gennip	
Contact Hours / Week x/x/x/x	0/4/0/0 hc; 0/4/0/0 wc	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Analysis 1 (AM1040), Linear Algebra 1 (AM1030) Other topics from first year courses are needed as well, such as: -series from Mathematical Structures (AM1010); -complex numbers from Kaleidoscope (AM1020); -partial derivatives, level set, and conservative vector fields from Analysis 2 (AM1070).	
Course Contents	This course is about ordinary differential equations. For first order equations we will treat linear, separable, and exact equations. We will prove existence and uniqueness of solutions, using Picard iterations. For second order equations we will treat linear equations, power series solutions, and the method of Frobenius. The Laplace transform will also be treated, as will systems of linear differential equations. Stability and phase plane analysis will play an important role.	
Study Goals	After finalizing the course, you should be able to use the theory of ordinary differential equations (ODEs) to study (systems of) differential equations. In particular you should be able to: 1. Solve linear (systems of) ODEs. 2. Solve separable ODEs and exact ODEs. 3. Determine the stability of solutions of ODEs. 4. Prove or disprove local existence and uniqueness of solutions. 5. Solve second order ODEs using power series and Frobenius theory. 6. Solve ODEs using Laplace transforms. 7. Determine features (like periodic behaviour, asymptotic behaviour, boundedness) of (solutions of) nonlinear (systems of) ODEs and sketch their phase planes. 8. Use ideas and techniques from the proofs in the course to prove similar and related results.	
Education Method	Lectures (4 hours a week) and exercise classes (4 hours a week)	
Literature and Study Materials	The book that will be used in this course is Differential Equations and their Applications, Fourth Edition. Martin Braun, Springer, 1993, ISBN: 0-387-97894-1 Additional materials will be provided during the course.	
Assessment	There will be a single written final exam (100%). The resit consists of a single written exam (100%). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You may use a graphical calculator at the exam. If necessary tables of Laplace transforms will be distributed.	
Co-Instructor	Dr. J.L.A. Dubbeldam	

AM2040	Complex Function Theory	6
Responsible Instructor	Dr.ir. W.G.M. Groenevelt	
Contact Hours / Week x/x/x/x	0/0/0/4 hc: 0/0/0/4 wc	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	AM1010 Mathematical Structures, AM1020 Kaleidoscope (Complex Numbers), AM1040 Analysis 1, AM1070 Analysis 2	
Course Contents	This course offers an introduction to the theory of analytic functions of one variable. An analytic function is a complex-valued function defined on an open set in the complex plane that is complex differentiable on its domain. Unlike functions in a real variable, the mere existence of a complex derivative has strong implications for the properties of the function. At the center of the course is the Cauchy integral formula for analytic functions. This formula expresses the value of an analytic function inside a circle in terms of the values of the function on the circle. The integral formula allows us to derive numerous beautiful properties of analytic functions. The general goals of the course are: - Being able to extend the basic notions from real-valued function of a real variable to complex-valued functions of a complex variable and having an understanding of the similarities and differences between real and complex analysis. - Acquiring analytic skills that enable one to choose an applicable methods from Complex Analysis for problems from various fields such as mathematical physics, system theory, signal analysis and the skills to turn this choice into a solution of the given problem. The following topics will be covered: 1. The basic complex function and their properties: the exponential function, trigonometric functions and logarithms 2. Analytic functions, the Cauchy-Riemann equations, harmonic functions 3. The Cauchy integral formula with applications, such as Liouville's theorem and the maximum-modulus principle. 4. Power and Laurent Series 5. Classification of isolated singularities 6. The residue theorem with applications, such as the argument principle and Rouché's theorem.	
Study Goals	By the end of the course you should be able to 1. perform basic mathematical operations (arithmetics, powers, roots) with complex numbers in Cartesian and polar forms, and solve elementary complex equations; 2. describe mapping behaviour of complex functions, including multi-valuedness and branching; 3. find the Taylor and Laurent series of a function and determine its domain of convergence; 4. determine the isolated singularities of a function, as well as the type of the singularities; 5. evaluate contour integrals using parametrization, primitive functions, Cauchys integral formula and the Residue theorem; 6. state and apply fundamental properties of analytic functions, including the Cauchy-Riemann equations, Liouvilles theorem, the maximum-modulus principle, the Identity theorem, Rouchés theorem; 7. apply various techniques to evaluate (improper) integrals, including contour integration, ML-estimates, Jordan's lemma, shrinking path lemma; 8. explain concepts, state and prove theorems and properties involving the above topics.	
Education Method	Lectures and exercise classes	
Literature and Study Materials	Nakhlé H. Asmar, Loukas Grafakos - Complex Analysis with applications, Undergraduate Texts in Mathematics. Springer, 2018, ISBN 978-3-319-94062-5. A digital version of this book is accessible online for TU-students via the TU library.	
Assessment	Written exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	The overall grade is the grade for the exam or for the resit.	
Co-Instructor	Dr. J.A.M. de Groot	

AM2050-A	Modelling 2A	3
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	a. Required: Introduction to statistics b. Strongly advised: Introduction to probability	
Course Contents	This modelling course focusses on applications of probability and statistics. Each group will work on an assignment handed out at the beginning of the course. Supervision consists of weekly meetings with the group's supervisor. Disclaimer: the study guide information may change depending on the developments around the coronavirus.	
Study Goals	By the end of the course, you should be able a. Apply standard statistical techniques to solve real-world like data-driven problems. b. Apply more advanced statistical techniques and compare those with the standard employed techniques. c. Analyse the results obtained from the standard and advanced statistical techniques and evaluate the performance of the methods. d. Introduce and motivate the choice of the statistical methods, and present and discuss results of the (performance) analysis in a report. e. Provide recommendations for the stakeholders course of action. f. Breakdown the problem in multiple tasks and distribute the work within the team. g. Present your results in an engaging and interactive way in a video.	
Education Method	-After an introductory meeting in the first week, there will be weekly meetings of each group with its supervisor. - Groups will produce midterm reports and receive feedback from supervisors. - At the end of the course, groups hand in the final report and summarize their findings with a short video.	
Literature and Study Materials	See Brightspace	
Assessment	Report (80%) Video (20%) In case of an insufficient final result, repair options may exist in accordance with Article 2 Examinationequirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Based on the written report (80% of the final grade) and video with findings (20% of the final grade)	
Co-Instructor	Dr. D. Kurowicka	

AM2050-B	Modelling 2B	3
Responsible Instructor	Dr. H.M. Schuttelaars	
Instructor	Dr.ir. E.G. Rens	
Instructor	Dr. A. Geyer	
Instructor	Dr.ir. Y.M. Dijkstra	
Contact Hours / Week x/x/x/x	0/0/0/4 prac	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	n.a.	
Course Contents	The course focusses on modelling a practical mathematical physics problem.	
Study Goals	At the end of the course, the student 1. is able to develop a simple mathematical model for a given practical problem. 2. is able to apply these methods and techniques to a practical modelling problem. 3. is able to acquire additional knowledge that is relevant for solving a practical problem. 4. is able to apply the mathematical modelling cycle and comment on the usefulness and weaknesses of mathematical models. 5. is able to determine the consistency and the validity of a mathematical model by analysis of simple test cases. 6. is able to communicate with the problem owner (supervisor) to get relevant information necessary for solving the practical problem and can deal with the uncertainties in this information. 7. is able to determine the different sub problems and is able to solve the complete problem as a team. 8. is able to present the results of the project through an oral presentation and by writing a report. 9. is able to use tools as programming languages, mathematical software and application software. 10. has a better overview of the applicability of mathematics.	
Education Method	Working on the project in a group of studenten with weekly contact with the supervisor of the project	
Literature and Study Materials	n.a.	
Assessment	The assessment is based on the assessment rubric. Here we have the following aspects: - Research : Quality of work done in the project (30%) - Process: The way the work has been carried out (30%) - The report (30%) - The presentation (10%) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	The study guide information may change depending on the developments around the coronavirus	
Judgement	Four grades for the aspects: - Research (30%) - Process (30%) - Report (30%) - Presentation (10%) Final grade is a weighted average of these four grades	
Co-Instructor	Dr. A. Geyer	

AM2060	Numerical Methods 1	6
Responsible Instructor	mr.ir.drs. V.N.S.R. Dwarka	
Contact Hours / Week x/x/x/x	0/0/2/2 hc; 0/0/2/4 wc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	Further numerical courses, modelling and BSc-project	
Expected prior knowledge	Analysis I and II, Linear Algebra I and II	
Summary	Introduction to numerical mathematics, rounding/truncation errors, interpolation, numerical differentiation, numerical time integration for initial value problems, boundary value problems (finite differences), heat equation, numerical integration (quadrature), non-linear equations.	
Course Contents	Numerical mathematics is the study of methods that give numerical approximations to mathematical problems. In most cases, the exact solution cannot be obtained (easily). During this course, numerical methods are treated for the solution of ordinary differential equations, and a relatively simple partial differential equation (the time-dependent heat equation). Further several numerical methods are treated which can also be applied to problems that do not involve differential equations. Such methods are: interpolation, numerical integration, and solution strategies for non-linear equations. The use of numerical software (MATLAB/Python) is an essential part of the course material.	
Study Goals	After having completed the course, the student will be able to design, analyse and implement numerical methods for ordinary differential equations. At the end of the coarse, the student will be able to: -design numerical methods for ordinary differential equations -analyse properties of the numerical methods (for example stability and consistency) -understand the numerical treatment of time-dependent and stationary ordinary differential equations and/or a combination of both -distinguish between the numerical treatment of a linear and non-linear problem -implement the numerical methods (for example using python or matlab)	
Education Method	Two hours of lectures and an afternoon of lab work/instruction per week: -During the instruction hour, you will be able to work on homework sets. These homework sets are for practice purposes and are not-graded. -During the lab instruction, you will be able to work on the lab assignment (programming).	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations, C. Vuik, F.J. Vermolen, M.B. van Gijzen, en M.J. Vuik (DAP / VSSD, 2015)	
Assessment	Lab assignment (pass/fail which grants access to exam) and a written test. Only a pass from the previous study study year will be valid. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen, non-programmable calculator and workpaper (which will be provided on the spot)	
Judgement	The final grade is determined as follows: 100% written test. The final grade will only be awarded provided the lab report is satisfactory.	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM2070	Partial Differential Equations	6
Responsible Instructor	Dr. H.M. Schuttelaars	
Contact Hours / Week x/x/x/x	0/0/4/4 hc	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course Ordinary Differential Equations AM2030	
Course Contents	Quasi-linear first order PDEs, and a simple model describing traffic flow.Types of second order PDEs : parabolic, elliptic, and hyperbolic PDEs. Derivation of the wave equation. Initial and initial-boundary value problems. Waves and reflctions. Fourier series and the method of separation of variables. Sturm-Liouville problems. derivation of the heat equation. Boundary value problems. Delta functions and distributions. Green's functions for heat, wave, and Laplace equations. Fourier and Laplace transform methods.	
Study Goals	At the end of the course, the student should be able to: 1. solve initial value problems and initial-boundary value problems for wave, transport, and heat equations. 2. solve boundary value problems for Laplace and Helmholz equations. 3. make the right choice for the method to solve the problems listed in 1 and 2. 4. give a physical interpretation of the the constructed solutions. 5. visualize the results by using the formula manipulation package MAXIMA.	
Education Method	lecture, exercise class, and practical work	
Literature and Study Materials	R.Haberman, Applied Partial Differential Equations, Fifth edition, Pearson Prentice Hall, New Jersey, 2013, ISBN-13: 9780321828972	
Assessment	Home-work exercises, practical work, and a written exam The content of the courses WI3150TU + WI3151TU is the same as course AM2070. However, the assessment of AM2070 is not the same as for the courses WI3150TU and WI3151TU. Therefore the courses WI3150TU + WI3151TU cannot be exchanged at will: you must first submit a request to the Board of Examiners. "The final mark for the course is the mark for the exam, but it is only valid when all practical work have been completed with a pass. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	pen, pencil, passer, geo-triangle (tabels will be handed out during the exam)	
Special Information	the study guide information may change depending on the developments around the coronavirus.	
Judgement	Final grading. 1) The practical work is mandatory, and as long as the practical work is not completed no final grading will be given. 2) Twice per year a written exam will be held. For the written exam in the fourth teaching quarter the following rules hold. The final grading is equal to the maximum of { the written exam grading, the average of the written exam grading and the home-work grading (where the exam grading has to be at least 5.0)}.For the written exam in the other period the final grading is equal to the written exam grading.	
Co-Instructor	Ir. W. van Horssen	

AM2080	Introduction to Statistics	6
Responsible Instructor	Prof.dr.ir. G. Jongbloed	
Instructor	Dr. A.F.F. Derumigny	
Contact Hours / Week x/x/x/x	4/0/0/0 hc; 4/0/0/0 wc	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Introduction to Probability (first year probability course)	
Course Contents	Statistical Models; Descriptive Statistics, Estimation theory; Testing theory; Confidence regions; Regression	
Study Goals	By the end of this course: 1. You know the concept of statistical model and you are able to formulate such a model for standard datasets. 2. You know several numerical and graphical summaries of a dataset and their relation with the probability distribution in the statistical model for the corresponding dataset and you are able to produce them with the statistical package R. 3. You know the concepts of Mean Squared Error and unbiasedness for an estimator and you are able to determine them for simple estimators. 4. You know the Maximum Likelihood, Method of Moments, and Bayesian approach towards estimation and you are able to construct estimators according to these principles and compare their performance either by hand or by means of a simulation with the package R. 5. You know the basic concepts involved in hypothesis testing and you are able to setup and conduct a statistical test in simple statistical models. 6. You know the setup of the Student-t test for one sample, for two paired samples, and two independent samples, and you are able to perform these tests either by hand or by means of the package R. 7. You know the setup of likelihood ratio tests and you are able to perform these tests either by hand or by means of the package R. 8.You know the concept of confidence interval and you are able to construct these intervals from pivots or near-pivots for simple statistical models. 9. You know the formulation of the linear regression model and the maximum likelihood estimators for the unknown parameters and you can fit these models by means of the statistical package R. 10. You know several concepts to check the assumptions and assess the fit of a linear regression model and you are able to determine these by means of the package R. 11. You know the content of important results, such as Theorem 4.29 and its proof, Theorem 4.43 and Theorem 5.9. 12. You know the definition of score function and Fisher information and their properties formulated in Lemma 5.10.	
Education Method	Lectures, exercise classes, and computer assignments.	
Computer Use	Statistical analyses are performed by means of the (free-ware) package R. See www.r-project.org	
Literature and Study Materials	"An Introduction to Mathematical Statistics", 1st edition. Fetsje Bijma, Marianne Jonker and Aad van der Vaart. Slides of the lectures are made available through Brightspace. Weekly written and/or computer assignments Statistical Software package R and the program R-studio	
Assessment	Mandatory R-assignments that have to be completed with a PASS. Written exam in week 10 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	3 hour written exam	
Permitted Materials during Tests	When the exam is held on campus, then the exam is a "closed book" exam. You are only allowed to use sheets with information on probability distributions and R-commands. You are not allowed to use any books or notes. When the exam is held online, then book and personal notes are permitted. NOT permitted is contact with third parties or getting information from the internet during the exam.	
Judgement	Grade of the written exam (or the resit). The course is completed only with a sufficient grade for the written exam and if all R-assignments have been completed with a PASS	
Co-Instructor	Dr. rer. nat. F. Mies	

AM2090	Real Analysis	6
Responsible Instructor	Dr. Ir. E. Lorist	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Required for	All subsequent courses in analysis and probability theory.	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Linear Algebra 1.	
Course Contents	The course consists of two parts. The first part of the course is about metric spaces. Metric spaces are a generalization of, e.g., the Euclidean space and are needed in applications to various other mathematical fields. Common examples are: - The space of continuous functions (Differential equations) - The space of square integrable functions (Partial differential equations, Numerical analysis, Probability theory, Quantum mechanics, Graph theory, etc.) - The space of summable sequences (Machine learning) You will learn about metric spaces during the course. Among other things, we will discuss: convergent sequences, open and closed sets, continuity and homeomorphisms, total boundedness and completeness, compactness, uniform convergence and spaces of continuous functions. The second part of the course is about measure and integration theory. This theory is at the foundation of a large part of modern mathematics. It is used in Fourier analysis, (partial) differential equations, probability theory and numerical analysis. In measure theory, we study how to measure a set in a fair way. We will be able calculate volumes and areas of sets that cannot be treated with the Riemann integration theory from Analysis 1 and 2. This will lead us to a larger class of functions which we can integrate and very strong convergence theorems. We will end the second part of the course with an application to Fourier series.	
Study Goals	At the end of the course, you should be able to write rigorous mathematical proofs in the context of metric spaces and Lebesgue integration theory. The first quarter focusses on metric spaces and the second quarter on Lebesgue integration theory. By the end of the first quarter, you should be able to LO1.1: Formulate the definition of the main notions of the theory of metric spaces and explain the connection to their counterparts in the real number line. LO1.2: Prove the characterizations of convergent sequences, open and closed sets and continuous functions treated during the course. LO1.3: Use the definitions of totally bounded, complete and compact metric spaces to prove properties of such spaces. LO1.4: Apply the definition of pointwise and uniform convergence to show results for sequences of continuous functions. By the end of the second quarter, you should be able to LO2.1: Apply the definition of a measure (in particular, the Lebesgue measure) and the notions it builds upon to prove basic properties. LO2.2: Use the construction of the Lebesgue integral for measurable functions and its connection to the Riemann integral for continuous functions to prove results for integrals. LO2.3: Prove and apply the three main convergence theorems (the monotone convergence theorem, Fatou's lemma and the dominated convergence theorem). LO2.4: Perform manipulations in L^p spaces, like applying Minkowski's and Holder's inequality and using completeness.	
Education Method	Lectures + Exercise classes	
Literature and Study Materials	Real analysis, N. L. Carothers, Cambridge University Press, Cambridge, 2000, ISBN 0-521-49756-5	
Assessment	Two 2-hour exams, one after each quarter. The final grade is determined as the average of these grades. A 3-hour resit about both parts of the course, which directly determines the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Pen and paper	
Co-Instructor	Prof.dr.ir. M.C. Veraar	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

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AM2510	Decision Theory	6
Responsible Instructor	G.F. Nane	
Contact Hours / Week x/x/x/x	0/0/4/0 colstr; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Probability theory and Statistics	
Course Contents	The course covers different aspects of the following subjects: Forecasting Inventory Management Judgmental Forecasting Decision trees Influence diagrams Bayesian Networks	
Study Goals	1. Learn how to model and solve real-life decision problems. 2. Understand and master the theoretical concepts covered during the lectures or used during the projects. 3. Understand the methods taught during lectures in detail, as well as the assumptions and motivation. 4. Apply statistical methods to real life situations 5. Provide support in decision-making problems. 6. Are able to master basic software statistical tools and to find appropriate functions to use during projects. 7. Are able to communicate the results to the problem owner and can justify/motivate the undertaken steps.	
Education Method	Lectures & Projects Expected workload 1. Attending lectures and practicum: 72 hours 2. Working for the project assignments (apart from the practicum time): 80 hours 3. Preparing for the oral exam: 10 hours 4. Oral exam: 1 hour 5. Preparing lectures, homeworks, correcting other assignments: 5 hours	
Computer Use	R, Python, Matlab, Excel	
Literature and Study Materials	Lecture slides and notes (diktaat), online videos and materials (from MOOC)	
Practical Guide	4 project assignments Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Books	additional (not required) reading: "Decision Systems for Inventory Managment and Production Planning" E.A.Silver "Experts in Uncertainty" R.M Cooke	
Reader	Basic Concepts of Applied Decision Theory (diktaat)	
Assessment	Reports of projects & Oral exam Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Reports for 4 projects are judged from the perspective of : - correctness of analysis - clarity of presentation - innovativeness of proposed method Oral exam: Discussion of theory used in the projects, results obtained and conclusions drawn. - each report is graded and feedback is provided. For the first two reports, the students can improve their grade by 0.5 points, if they address all the comments provided in the feedback. - during the oral exam, the grade of each project is updated based on the answers of students during the oral exam.	
Co-Instructor	Dr. D. Kurowicka	

AM2520-H	History of Mathematics	6
Responsible Instructor	Dr. R.J. Fokkink	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	No specific prior knowledge, other than the content of the first three semesters year of the curriculum Applied Mathematics, is required. We do explicitly expect a wide interest in mathematics, its history, and its philosophy.	
Summary	Disclaimer: the study guide information may change depending on the developments around the coronavirus. Only on September 1st will it be in its final form.	
Course Contents	This course gives an overview of parts of the history of mathematics. Which parts may to some extent be dictated by the interest of the students, although certain `standard' developments will be treated regardless. Among the `fixed' topics are - developments in Egyptian and Mesopotamian mathematics - Greek mathematics - mathematics in the Indian and Arab worlds - how certain notions developed through time: calculus, algebra, geometry - the foundations of mathematics, then and now	
Study Goals	At the end of the course: 1. the student knows some of the major developments in the history of mathematics, in particular the mathematics that forms the core of the curriculum at the TU Delft; 2. the student is able to study, independently, the historical apsects of a mathematical subject or area and report on this study; 3. the student is able to study a historical subject or paper and report on this orally and in writing.	
Education Method	A mix of lectures, discussions, presentations by students and written work (an essay on a subject of the student's choice). Major emphasis is on active participation. Every student will give at least two oral presentations on some subject (prepared in advance). Attendance is compulsory.	
Literature and Study Materials	There is no specific text that will be used. Throughout the course we create a reader by offering reading material on Brightspace. Thanks to world-wide digitization efforts we can use many original texts. Some books can be used as background material. There will list of recommended books on Brightspace in due course.	
Assessment	There is no written exam. The grade will be based on short quizzes about the reading material, on the oral presentations, and on the essay. These will get their own grades that are weighed as follows: essay 40%, presentations 40%, quizzes 20%. The number of participants may necessitate some deviation from the rules. If that happens then this will be made known via Brightspace. For the whole course attendance is compulsory. The precise regulations will be available on Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	BSC AM students can choose either AM2520-H or AM2520-P. It is not allowed to enter both courses in the individual study programme.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM2520-P	Philosophy of Mathematics	6
Responsible Instructor	Prof.dr. J.M.A.M. van Neerven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge, other than that covered in the first three semesters of the TW study, is not required. A broad interest in mathematics, science and philosophy in general and an open attitude are, however, required.	
Course Contents	In this course we investigate what mathematics is and how it relates to other (exact) fields of science. Central is "the appicability of mathematics as a philosophical problem": how can mathematics, as a formal language, have applications in the world around us? The following topics are covered: - what is knowledge? - what is science? - what is math? - foundations of mathematics: logic, truth, proof - main currents in the philosophy of mathematics - Wigner's "unreasonable effectiveness of mathematics" - the mathematical description of external reality: philosophical aspects of space, time and matter in classical mechanics, relativity theory, quantum mechanics and cosmology - mathematical aspects of determinism and causality - mathematical aspects of free will and self-awareness All these topics will be dealt with in their historical context. We will not only discuss current views and debates, but also the evolution of thinking on these subjects over the centuries, starting in Antiquity.	
Study Goals	After the course: 1. the student is familiar with the main points of the philosophy of mathematics and its applications in the natural sciences; 3. the student is able to independently study the philosophical aspects of a mathematical subject or sub-field and to report on this; 4. the student is able to explore and present a philosophical subject or article orally and in writing.	
Education Method	A mix of lectures, discussions, and student presentations. Great emphasis is placed on active student participation. Every student gives at least two oral presentations on a pre-prepared topic. Attendance is compulsory for this entire course.	
Literature and Study Materials	A reader is built up by offering weekly reading material via Brightspace. In the background we use some books, the titles of which will be announced via Brightspace as well.	
Assessment	The final grade is the weighted average of the grades for the presentations during the course and the grade for the final exam. In the final (written) exam, the student will write an essay on a topic relating to the subject matter of the course. Further details will be offered via Brightspace. Attendance is compulsory for this entire course. The exact regulation can be found in the course description on Brightspace. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Exam Hours	A final written exam of 3 hours.	
Permitted Materials during Tests	Pen and paper	
Remarks	1) Only one of the two courses AM2520-H ofwel AM2520-P can be attended for ECTS credit points, but not both. 2) A cap of 20 students applies. If enrollments exceed this number, students will be selected on the basis of their motivation.	

AM2550	Advanced Statistics	6

Responsible Instructor	Dr. N. Parolya
Contact Hours / Week x/x/x/x	0/0/2/0 hc; 0/0/2/0 wc; 0/0/4/0 prac
Education Period	3
Start Education	3
Exam Period	3 4
Course Language	English
Expected prior knowledge	Introduction to Statistics TW2080; Introduction to Probability Theory TW1080; Linear Algebra TW1030, TW2011
Course Contents	Mathematical foundations of statistical methods and models used for multivariate data analysis with linear models. Application-oriented component for analyzing and interpreting the data with "R". Short outline of the topics: 1. Multiple linear regression (OLS and GLS) 2. ANOVA (one-way and two-way) 3. Generalized Linear Models (logit and probit) 4. Generalized Least Squares (heteroscedasticity and white errors)
Study Goals	After successful completion of this course, the student is able to LO1: describe the definition of the multiple linear regression model; LO2: create a multivariate linear model, which fits the best the underlying data set; a: identify the relevant explanatory variables; b: find the least squares estimator; c: interpret the estimated coefficients; d: provide the corresponding confidence and prediction intervals; LO3: analyze the ANOVA model; a: recognize ANOVA from a general class of linear models; b: estimate its unknown parameters; c: provide the valid statistical inference of the model; d: interpret the final results in terms of mean profiles; LO4: analyze the Generalized Linear Model (GLM); a: define the GLM; b: estimate the GLM; c: provide a valid asymptotic inference; d: interpret the obtained results in terms of odds; LO5: handle the general linear models with heteroscedasticity and autocorrelation; a: verify whether the classical assumption of homoscedasticity is valid; b: apply weighted least squares method efficiently; c: interpret the white standard errors; LO06: give proofs of the main theoretical statements about the statistical properties of the estimators. LO07: apply the learned methods in practice using the statistical software R. (LO stays for learning objective)
Education Method	Lectures, tutorials, computer sessions
Literature and Study Materials	Syllabus via Brightspace, notes on Brightspace and from the lecture
Books	Applied Regression Analysis and Generalized Linear Models by John Fox, SAGE Publications 2016 Stock, J. H. and M. W. Watson (2012). Introduction to Econometrics, Pearson.
Assessment	Written or oral (in case the number of students <10) examination (70% of the final grade). Resit as written or oral examination (70% of the final grade), again depending on the number of students. Weekly compulsory computer exercises (30% of the final mark). In case of an insufficient final result, repair options may exist for the exam or computer assignments in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Permitted Materials during Tests	A graphical calculator A sheet of A4 paper, handwritten on both sides.
Judgement	Computer assignments: 5 best of 6 assignments amount to a grade: C Exam: grade T. Final grade= 0.3*C + 0.7*T Resit: grade R. Final grade for resit= 0.3*C + 0.7*R The minimum passing grade for the C, T, and R exams is 5.0.
Co-Instructor	Dr. J. Söhl

AM2560	Applied Algebra: Codes	6
Responsible Instructor	Dr. J.A.M. de Groot	
Contact Hours / Week x/x/x/x	0/0/4/0 hc 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch (on request English)	
Expected prior knowledge	Linear Algebra 1, Algebra 1	
Course Contents	The goal of this course is to show some practical applications of algebra. The course will have the following structure: -Introduction to coding theory (hamming distance, linear codes, parity check matrices and generator matrices, syndrome decoding) -An introduction to finite fields -Continuation of coding theory (cyclic codes, BCH codes, Reed Solomon codes) The dutch version of the course contents is much much more extended.	
Study Goals	Understanding and being able to apply the materials and techniques of this course. Furthermore, the ability to apply experience and knowledge to standard as well as non-standard problems.	
Education Method	Four hours lecture and four hours exercise class per week. A lot of material (for example the lecture notes) is in dutch and will not be translated to english when the course is given.	
Literature and Study Materials	Coding Theory and Cryptography, the essentials, 2nd edition, written by D.R. Hankerson, D.G. Hoffman, and five others (published by CRC Press). Lecture slides (Brightspace) Extra exercises (Brightspace) An additional reader containing some extra theory and the extra extra exercises.	
Assessment	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	A three hour, written, final exam at the end of the third quarter, with a resit exam in the fourth quarter.	
Co-Instructor	Dr.ir. J.H. Weber	

AM2570	Markov Processes	6
Responsible Instructor	Dr. C. Kraaikamp	
Contact Hours / Week x/x/x/x	0/0/6/0 hc; 0/0/2/0 instr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	The expected prior knowledge is the content of the first year course "Introduction to Probability Theory" (AM1080). Active knowledge of the students of the two partition theorems, of probability and moment generating functions and of condional probabilities and conditional expectations is assumed.	
Course Contents	This is an introductory course on Markov chains, where time-discrete and continuous time Markov chains will be introduced, and their most fundamental basic properties studied. Properties as the matrix of transition probabilities (and in the continuous case the Q-matrix), stationary distribution, ergodicity of the Markov chain, recurrence and transience of states will be discussed. Very many examples will be presented to illustrate these notions.	
Study Goals	After this course you can work with the basic properties of time discrete and continuous time Markov chains. You can determine the matrix of transition probabilities (and in the continuous case also the Q-matrix), you can determine the invariant measure in the instances it exists and derive from this conclusion about the system you are studying, in particular the long-term behavior of this system. You should be able to use the theorems discussed in class effectively to solve the problems, and the various notions introduced in class (such as irrreducibility and (a)periodicity) will be a part of your tool set.	
Education Method	This course will be taught in a standard classroom setting, with exercises hours. You have to prepare in advance for the exercise hours, and then during these exercise hour the exercise will be solved by you and the lecturer in an interactive way.	
Literature and Study Materials	In this course the same book (by Grimmett and Welsh) will be used as in AM1080. There is also a reader, which will be made available on the Brightspace page of this course. Other literature mentioned in class can always be found on-line in the library of the TU Delft, and is available for the students.	
Assessment	There will be weekly assignments, but the final grade is determined only by the written exam. The weekly assigments can be handed in for comments, but no grade will be given. The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Prof.dr. F.H.J. Redig	

AM2580	Mathematical Models in Biology	6
Responsible Instructor	Dipl.-Math. H. Yoldas	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course introduces prominent mathematical models and modeling approaches in biology ranging from population dynamics, cell signaling, reaction kinetics, epidemiology, and neuroscience. We will cover a variety of discrete (agent-based models, difference equations) and continuum (ordinary and partial differential equations (ODEs, PDEs)) mathematical models. We will employ diverse mathematical techniques including dimensional analysis, non-dimensionalization, stability analysis, parameter estimation, and numerical analysis to study these models. A special attention will be paid to the biological interpretation of the models (modeling parameters and the relevance of the modeling assumptions), their applicability to the real world and the implications of the mathematical results. Several computational tools will be introduced and applied during the lab hours. The lectures together with accompanying instructions and lab work will provide a profound introduction to mathematical biology.	
Study Goals	After successfully completing the course, you should be able to: LO1: Classify mathematical models correctly, distinguishing between deterministic and stochastic, discrete and continuum, ODE and PDE, as well as microscopic, mesoscopic, and macroscopic models. LO2: Interpret the variables, parameters, and equations within a given biological context for the models outlined in LO1. LO3: Justify the suitability of specific types of models identified in LO1 within a given biological context. LO4: Apply appropriate (analytical and numerical) mathematical techniques to analyze the mathematical models described in LO1 effectively. LO5: Explain the biological implications of the results obtained through the analysis outlined in LO4.	
Education Method	Lectures, instructions, and computer lab sessions.	
Computer Use	Python (or Matlab), some cell-based modelling software packages.	
Literature and Study Materials	Lecture slides Some chapters from extra reference books provided on the Brightspace page. Two main reference books are Mathematical Models in Biology by Leah Edelstein-Keshet and Mathematical Biology by James D. Murray.	
Books	Two main reference books are Mathematical Models in Biology by Leah Edelstein-Keshet and Mathematical Biology by James D. Murray.	
Assessment	The final grade of the course consists of the following components: - Assignments: 30% - Written exam: 70% Final grade calculation: max{5.8, (0.3 * Grade obtained from the assignments + 0.7 * Written exam grade)} The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Pen and paper	
Co-Instructor	Dr.ir. E.G. Rens	

3e jaar BSc TW 2024

Program Coordinator	Bachelorcoordinator: Bart van den Dries BC-TW@TUDelft.nl
Prerequisites	Be aware! The entry requirements have been recorded in the OER. If there is a difference between what is listed here and what had been recorded in the OER, the content of the OER is leading. To participate in the course AM3000 Bachelorproject, it is mandatory to have finished all courses of the first year and at least 40 EC of the second year and the second semester of the third year.
Introduction 1	Enrollment: For the (partial-)exams of all courses, you have to register via Brightspace/OSIRIS ultimately one week before the day of the exam. For more information, look at https://www.tudelft.nl/en/student/education/examinations/. Do you not want to participate in a study component for which you have enrolled? Please unroll yourself via Brightspace/OSIRIS!
Introduction 2	In the third year you will do a minor. For more information, see http://minors.tudelft.nl. If want to do a master after your bachelor, the now is the time to inform yourself. For this pay attention to the announcements on Brightspace about the Master Events!

AM3001	Bachelor Project	15
Responsible Instructor	Dr.ir. J.T. van Essen	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	Dutch English	

AM3010	Bachelor Colloquium	3
Responsible Instructor	Dr. J.G. Spandaw	
Responsible Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	Check the Dutch version	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

Minor (30 EC)	
Director of Education	Bachelorcoordinator: Bart van den Dries BC-TW@TUDelft.nl

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Applied Mathematics**

keuzevakken 3e jaar BSc TW 2024

AM3500	Mathematics Seminar	6
Responsible Instructor	Dr. A. Agresti	
Responsible Instructor	F. Germ	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	This course has a different topic every year. For the academic year 2024/2025 this course is organized by the Analysis group and the topic is dynamical systems and their application in climate science. A textbook and other reading material will be made available to students, from which the talks for this course can be prepared. This course will focus on the mathematical theory of (nonlinear) dynamical systems and their applications to certain important models in climate science. In particular this course will consist of two parts: - in the first part we introduce and develop some theoretical notions of dynamical systems. This will include a review of methods in ordinary differential equation (ODEs) as well as some more advanced concepts with regards to their analysis. We will also introduce elements of stochastic calculus to analyze random dynamical systems. - in the second part of the course we will apply our theory to models from climate science. This includes the Climate Modelling Hierarchy, the North Atlantic Oscillations, the El Nino Variability, etc. We may also include research papers to give insight into modern research on the topics.	
Study Goals	See the Brightspace page of this course.	
Education Method	The course mainly consists of lectures delivered by the students. The role of the instructors is to coach the students, provide feedback on the lectures, as well as to give additional perspectives on the course material. Necessary prerequisites beyond the courses from the first two years may be taught by the instructors depending on the background of the students. The course material is divided among the students in the beginning of the course. The students prepare lectures about the assigned material including detailed proofs of relevant results.	
Literature and Study Materials	The course treats a selection of chapters from the following book (available as hardcopy via the library, or provided as copy in the course) Nonlinear Climate Dynamics by Henk A. Dijkstra, DOI: https://doi.org/10.1017/CBO9781139034135.001 Publisher: Cambridge University Press Print publication year: 2013 We will also provide supplementary material to cover theoretical aspects of dynamical systems.	
Prerequisites	Year 1: Mathematical Structures (AM1010), Analysis 1 and 2 (AM1040 and AM1070), Linear Algebra I (AM1030) Year 2: Ordinary Differential Equations (AM2030), and Real Analysis (AM2090), Linear Algebra II (AM2010)	
Assessment	The students' lectures are graded (70% of the final grade) and the course is concluded with either an oral exam or a group presentation of a modern research paper (30% of the final grade). In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination Requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Tags	Intensive Mathematics	
maximum aantal deelnemers	Due the special nature of the course (student lectures) there might be a limitation in the number of participants. If the number of enrolments is too high, priority will be given to the students from the Excellence Program. Secondary consideration will be given to performance on the prerequisite courses. Please enrol before the start of the course on the Brightspace page (once it is available) if you are planning to participate in this course. Enrolment after the first meeting is no longer possible.	

AM3510	Mathematical Physical Models	6
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	4/4/0/0 hc	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Ordinary Differential Equations AM2030, a first course in partial differential equations in which separation of variables is treated	
Course Contents	In this course we will develop a number of mathematical models for physical phenomena, especially in the context of continuum mechanics. All the models discussed are partial differential equations. The focus of the course is not on how to solve these equations, but how to derive them from the underlying physical laws, such as mass and energy conservation.	
Study Goals	The student is able to: 1. Formulate conservation laws 2. Use these laws to derive differential equations 3. Construct mathematical models for a number of physical phenomena related to continuum mechanics, such as displacement of material under loads, travelling of sound waves through a medium,.	
Education Method	Lectures	
Assessment	Homework exercises and an oral exam. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examinationequirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Judgement	Homework exercises and oral exam.	
Co-Instructor	Dr. J.L.A. Dubbeldam	

AM3530	Numerical Methods 2	6
Responsible Instructor	A. Palha da Silva Clérigo	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac/wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Numerical Analysis 1 (AM2060), Partial Differential Equations (AM2070)	
Course Contents	Finite differences, finite volumes in R^2, error analysis, discrete and continuous maximum principles, processing of boundary conditions, time integration.	
Study Goals	The focus of this course is on two of the three most common discretization techniques: the finite difference method and the finite volume method. Before introducing, deriving, and applying these methods, we start with the theoretical principles of partial differential equations (PDEs) that are necessary to analyze PDEs before discretizing them. We then introduce the two discretization techniques. Additionally, we introduce iterative methods to solve nonlinear PDEs, and finally, we will examine various numerical time integration methods, which we introduce using the heat equation and the wave equation. Specifically, by the end of the course, you should be able to: 1. Explain analytical properties of partial differential equations (PDEs) and the role of boundary and initial conditions that are important to obtain solutions to PDEs 2. Explain the properties, advantages and disadvantages of the finite difference method (FDM) and the finite volume method (FVM) to obtain approximate solutions to PDEs 3. Apply FDM and FVM to obtain approximate solutions to 1D and 2D systems of linear PDEs. 4. Apply two techniques to obtain solutions of non-linear PDEs, namely: the fixed-point scheme and Newtons method. 5. Implement numerical schemes for time-dependent PDE's on two specific equations: The diffusion and the wave equation. 6. Analyse the accuracy, stability, and convergence properties of all these above mentioned methods	
Education Method	Lectures, instructions + project	
Books	J. van Kan, A. Segal, F. Vermolen and H. Kraaijevanger. Numerical Methods for Partial Differential Equations, Delft Academic Press, Delft, 2019.	
Assessment	Project (50%) + oral exam (50%) (oral exam = 1 hour written exercises + 0.5 hour oral discussion of written exercises and project) In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Special Information	This course can not be taken in combination with TW3730TU Numerical Methods for Differential Equations	
Judgement	0.5*Project report + 0.5*Oral exam	
Co-Instructor	Prof.dr.ir. M.B. van Gijzen	

AM3540	Inverse Problems	6
Responsible Instructor	Dr. H.N. Kekkonen	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic linear algebra, matrix calculus, and Fourier transform. Familiarity with Matlab is useful but not mandatory.	
Course Contents	The goal of inverse problems is to determine the cause from the effect. For example, the cause can be a sharp image. The effect is a blurred and noisy image given as a result of a camera that is badly focused. Recovering the sharp image from the noisy measurement is a classic example of an inverse problem. Mathematically we can model blur in an image by convolution. The observed image is interpreted as a sharp image convolved with a kernel and corrupted with random noise. The related inverse problem of recovering a sharp image is called deconvolution. This inverse problem is ill-posed, meaning that the solution is highly sensitive to modelling errors and measurement noise. A stable solution can be produced using regularisation. Topics: Naïve inversion, (discrete) convolution, X-ray tomography, truncated singular value decomposition, Tikhonov and total variation regularisation, and short introduction to Bayesian approach to inverse problems.	
Study Goals	After this course you should Know how to model a measurement process as a matrix equation and detect if it leads to an ill-posed inverse problem. Understand why ill-posed problems require regularisation. Be able to design and implement a reconstruction method to recover the unknown object of interest from indirect and noisy measurements. Be able to solve discrete inverse problems using Matlab.	
Education Method	Lectures and exercise + (computer) labs	
Assessment	80% Final exam and 20% homework assignments. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr. Y. van Gennip	

AM3550	Graph Theory	6
Responsible Instructor	Dr. A. Bishnoi	
Co-responsible for assignments	A. Ravi	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra	
Course Contents	Graph theory is the main subject of this course. We will study a number of central topics from graph theory, including: 1. paths, cycles and trees 2. bipartite graphs 3. matching theory 4. graph colouring 5. planar graphs 6. spectral graph theory Our primary focus will be on theory and proofs, but throughout the course we will keep describing some real-life applications of the concepts introduced.	
Study Goals	By the end of the course, you should be able to: 1. Use the language of graph theory to formulate and solve various discrete problems. 2. Explain the proofs of some of the central theorems from graph theory (specified on brightspace). 3. Apply proof techniques like induction, double counting, and spectral methods (from linear algebra), to problems in graph theory.	
Education Method	Lectorials	
Books	Book: Graph Theory, by R. Diestel Book: Introduction to Graph Theory, D. West	
Assessment	Written examination The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	None	
Judgement	The exam grade is the final grade	
Co-Instructor	Dr. F.M. de Oliveira Filho	

AM3560	Advanced Probability	6
Responsible Instructor	J. Komjáthy	
Contact Hours / Week x/x/x/x	0/0/8/0 colstr	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The goal of the course is to extend the basic notions of probability with an analytic toolkit. Rigorous probability is closely related to real and complex analysis. We will explore these connections and learn how concepts from real analysis translate to probability and vice versa. In particular we will construct a general probability space and perform computations of expectations, and conditional expectation of random variables therein. We will treat different types of convergences of random variables, Borel-Cantelli Lemmas, Kolmogorov's 0-1 law. We will learn various analytic methods to prove convergence or random variables: characteristic functions, Fourier inversion, generating function. Laws of Large Numbers and general Central Limit Theorems will be discussed. If time permits, we will learn about stable distributions.	
Study Goals	At the end of the course, you are able to construct a general probability space. You are able to recognize, analyse, and construct (abstract) random variables as measurable functions, and sigma algebras. You are able to decide whether functions (random variables) and events are measurable. You are able to define and compute expectation and conditional expectation of arbitrary random variables, even when the conditioning is a 0 probability event and the random variables do not have a density. You are able to link the abstract definition of conditional expectation to earlier courses, e.g. conditional expectation in the setting if two dimensional random variables, and compute them using the combination of abstract theory and integration. You are able to compute the characteristic functions of well-known random variables, and use the characteristic function method to prove distributional identities. You are able to apply the tower rule to simplify long integrations into easy-to-compute formulas using conditional expectation. (which will be very useful in Stochastic Processes later on). You are able to invert the characteristic function of discrete random variables. You are able to apply Markov's and Chebyshev's inequality to bound the expectation and variance of sums of random variables. You are able to prove the Central Limit Theorem using characteristic function. You are able to prove Central Limit Theorem for non-iid random variables, you can apply and check Lindeberg's condition. Given a sequence of random variables, you are able to tell whether they convergence almost surely, in probability, in L^1 or in distribution. You are able to apply the Borel-Cantelli lemmas to prove that a given a sequence of random variables converges almost surely. You are able to combine analysis (of characteristic functions) to prove that a given a sequence of random variables converges in distribution. You are able to prove the Weak Law of Large Numbers using Chebyshev's inequality. You remember the limitations of the Law of Large Numbers, and recognize when it does not hold for a sequence of random variables. You are able to estimate the maximum of a collection of heavy tailed random variables using Markov's inequality and elementary computations.	
Education Method	The course content is based on the book "Probability Essentials" by J. Jacod and P. Protter. There will be 4h lectures and 2h exercises and 2h discussion of the exercises per week with homework (twice).	
Assessment	The assessment will consists of two homework assignments and one final written exam. One needs at least 5.0 in the assignments to be allowed to sit at the exam. The assignments do not count towards the resit. In case of an insufficient final result, repair options may exist in accordance with Article 2 Examinationrequirements, Clause 4, of the Implementation Regulations 2023-2024. The repair option is a resit exam that covers the material both in the homework assignments and the final exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	There is no additional material allowed during the exam. For the home assignments there is no restriction on the material allowed.	
Judgement	The grade will be calculated as follows: - 40% of the grade is based on home assignment (two assignments throughout the course); - 60% of the grade is based on the final written exam. The final exam must have a grade larger or equal to 5 to pass the course. - In case that the resit exam is taken, the final grade is given by 100% of the result on the resit exam; this overwrites the earlier attained result. (That is, it can be both lower or higher than the previously attained grade).	
Co-Instructor	Prof.dr. F.H.J. Redig	

AM3570	Fourier Analysis	6
Responsible Instructor	F. Bartolucci	
Contact Hours / Week x/x/x/x	4/4/0/0 colstr	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Before taking this course you should be familiar with the contents of the courses Real Analysis (AM2090) and Complex Function Theory (AM2040).	
Course Contents	The students will learn the fundamental concepts, results and methods of Fourier Analysis, on the torus and on the real line, and how to apply them to solve new exercises. The course will cover the following subjects: Fourier series: - Fourier series of periodic absolute integrable functions - Cesàro and Abel summability - Fourier series of periodic square integrable functions - pointwise convergence - applications. The Fourier transform on the real line: - Fourier transform of absolute integrable functions - convolutions and approximate identities - Fourier inversion theorem - Fourier transform of square integrable functions - Poisson summation formula - Paley-Wiener theorem (optional) - Shannon theorem (optional) - applications. The Fourier transform of tempered distributions (optional).	
Study Goals	At the end of the course, you should be able to: LO1) reproduce the notions and the statements of the theorems covered in the course LO2) reproduce the proofs of the results covered in the course LO3) apply theoretical results and proof techniques in solving new exercises.	
Education Method	The course consists of frontal lectures where the theory is explained and where applications are discussed. These are integrated with exercise classes.	
Reader	Lecture notes are available on Brightspace.	
Assessment	The exam consists in an oral exam. There will be weekly homework assignments. Students who acquire at least 70% of the points attainable by doing the weekly problem sheets are given an additive bonus of 0.5 on their final grade (e.g. grade 8 (without bonus) will be grade 8.5 (with bonus)). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	No materials are permitted.	
Co-Instructor	Prof.dr. J.M.A.M. van Neerven	

AM3580	Differential Geometry	6
Responsible Instructor	Dr. B. Janssens	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 wc	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Linear algebra I and II (TW1030 and AM2010), and Analysis I and II (AM1040, AM1070). Occasionally we will encounter metric spaces (Real Analysis, AM2090), groups (Algebra 1, AM1060) and analytic functions (Complex function theory, AM2040), but without the last three courses it will still be possible to follow the course. It is an advantage if you have taken a course on topology (e.g. AM3590), but this is not a prerequisite. The necessary material on topology is gathered in appendix A of the course notes. If you have not encountered topological spaces before, I recommend that you read at least A1 and A2 (2 pages) before the start of the course.	
Course Contents	Differential geometry is the study of smooth manifolds, such as spheres, tori, hyperbolae, etc. Since these spaces behave locally like a vector space, they are amenable to techniques from analysis and linear algebra. The focus of this course will be on Riemannian geometry, the study of metric spaces in a smooth context. At the end of the course, we will discuss the link between (pseudo)-Riemannian geometry and Einstein's theory of general relativity.	
Study Goals	At the end of this course, you should able to: 1) explain the basic notions of differential geometry, such as smooth manifolds, tangent bundles, Riemannian metrics, and geodesic paths. 2) calculate the Riemann curvature tensor for a given Riemannian metric, and apply this in the context of connections, curvature and geodesics. 3) explain and apply a number of basic theorems of differential geometry, such as Gauss' Theorema Egregium. 4) interpret and explain Einstein's equation for the gravitational field.	
Education Method	Lectures and exercise sessions	
Literature and Study Materials	Lecture notes, to appear on Brightspace.	
Assessment	Written examination and resit The resit consists of a single written exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	F. Bartolucci	

AM3590	Topology	6
Responsible Instructor	Dr. S.E. Zegers	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Mathematical Structures, Analysis 1 and 2, Real Analysis, and Algebra 1.	
Course Contents	The scope of the course is to provide a general introduction to topological spaces and their properties. Moreover, we will in the end of the course introduce the fundamental group which gives a tool of determining when two topological spaces are the same (homeomorphic). We will cover the following topics: - Definition of topological spaces and basis for a topology - Continuous functions - Compactness - Connectedness and path-connectedness - Separation properties: Hausdorff, normal, regular - The quotient, product, subspace and metric topology - The Tychonoff theorem - The Arzela-Ascoli theorem - The Stone-Weierstrass theorem - Homotopy of paths - The fundamental group, functionality - Brouwer fixed-point theorem for the disc	
Study Goals	-You can explain what general topological spaces are including their properties and illustrate this using various (well-known) examples described during the course. -You are able to explain the main notions that occur in all the topics of the course content. This concerns: compactness, continuity, separation properties, connectedness, the quotient, subspace, product and metric topology, homotopy of paths and the fundamental group. -From the most important theorems you are able to analyse the proofs. This concerns: Mutual relationships between separation properties and compactness, Tychonoff, Arzela-Ascoli and Stone-Weierstrass theorem, the fundamental group as a topological invariant, functionality and Brouwer fixed-point theorem for the disc.	
Education Method	Lectures and exercise classes.	
Literature and Study Materials	We will follow the book "Topology" by J. R. Munkres (ISBN 13: 978-1-292-02362-5, International edition) There will also be lecture notes as a supplement to the book.	
Assessment	Oral exam If the estimated number of students that takes the exam is above 15 then the exam may be written. This will be decided and announced during the first quarter of the course. The resit consists of an exam that counts for 100%. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
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